## 3.29 Fractional Ideals

**Definition 3.29.1**
*Let $A$ be an integral domain contained in its quotient field $K$. A non-zero $A$-submodule $I \subset K$ is called a* ***fractional ideal*** *if there exists $0 \neq a \in A$ such that $aI \subset A$.*

*We say a fractional ideal is* ***principal*** *if it is generated by some $x \in K$ as an $A$-module.*

**Proposition 3.29.2 (Fractional Ideals over a Noetherian Domain)**
*Let $A$ be a Noetherian integral domain. Then every fractional ideal of $A$ is a finitely generated $A$-module.*

*Proof.* Let $I$ be a fractional ideal and $0 \neq a \in A$ with $aI \subseteq A$. Then $aI$ is an ideal of $A$, hence finitely generated, say by $x_1, ..., x_n$. Multiplication by $a$ is an isomorphism $I \to aI$ of $A$-modules, so $I$ is generated by $a^{-1}x_1, ..., a^{-1}x_n$. ∎

**Proposition 3.29.3 (Operations on Fractional ideals)**
*Let $A$ be an integral domain and $I, J \subset K$ fractional ideals. Then the following constructions are also fractional ideals*

*a)* $I + J = \{i + j \mid i \in I j \in J\}$
*b)* $IJ = \left\{\sum_{k=1}^{n} i_k j_k \;\middle|\; i_k \in I j_k \in J\right\}$
*c)* $(I : J) = \{x \in K \mid xJ \subseteq I\}$

*Proof.* It is straightforward to prove that these constructions are again $A$-submodules of $K$.

Suppose that $aI \subseteq A$ and $bJ \subseteq A$, then $(ab)(I + J) \subseteq A$ and $(ab)IJ \subseteq A$. Similarly

$$(I : J)J \subseteq I \implies j(I : J) \subseteq I \implies abj(I : J) \subseteq A$$

for some $0 \neq j \in J$ ∎

**Proposition 3.29.4**
*Let $A$ be an integral domain. The collection of fractional ideals $\mathcal{I}_A$ forms a monoid under multiplication of ideals, with identity $A$.*

**Definition 3.29.5**
*Let $A$ be an integral domain and $I$ a fractional ideal. Then we say $I$ is* ***invertible*** *if there exists a fractional ideal $J$ such that $IJ = A$.*

**Proposition 3.29.6**
*Let $A$ be an integral domain and $I$ a fractional ideal. Then $I$ is invertible if and only if $I(A : I) = A$, in which case $(A : I)$ is the unique inverse.*

*Proof.* Uniqueness follows from general properties of monoids (...). If $I(A : I) = A$ then by definition $I$ is invertible. Conversely suppose $I$ is invertible, that is $IJ = A$. Then evidentally $J \subseteq (A : I)$ so $A = IJ \subseteq I(A : I) \subseteq A$. Therefore $(A : I)$ is an inverse for $I$ as required. ∎

**Corollary 3.29.7**
*Let $A$ be an integral domain. Then the invertible fractional ideals form an abelian group under multiplication.*

**Definition 3.29.8 (Localisation of Fractional Ideals)**
*Let $A$ be an integral domain contained in its quotient field $K$. Let $I$ be a fractional ideal and $S$ a multiplicatively closed subset of $A$. Then define*

$$S^{-1}I \coloneqq \left\{\frac{i}{s} \;\middle|\; i \in I s \in S\right\}$$

*If $\mathfrak{a}$ is an ideal then $S^{-1}\mathfrak{a}$ corresponds to the usual notion (3.5.17) and is equal to the extension of $\mathfrak{a}$ under the inclusion homomorphism $A \to S^{-1}A$.*

*If $\mathfrak{p}$ is a prime ideal of $A$ then as usual define*

$$I_{\mathfrak{p}} \coloneqq (A \setminus \mathfrak{p})^{-1} I$$

*and we may write*

$$\mathfrak{a}_{\mathfrak{p}} = \mathfrak{a} A_{\mathfrak{p}}$$

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**Proposition 3.29.9 (Localisation of Fractional Ideals)**
*Let $A$ be an integral domain contained in its quotient field $K$. Let $S$ be a multiplicatively closed subset of $A$ and $I$ a fractional ideal. Then regarding as subrings of $K$ then $S^{-1}I$ is a fractional ideal of $S^{-1}A$. Furthermore*

*a)* $S^{-1}(I+J) = S^{-1}I + S^{-1}J$
*b)* $S^{-1}(IJ) = (S^{-1}I)(S^{-1}J)$
*c)* $S^{-1}(I:J) = (S^{-1}I : S^{-1}J)$ *if $J$ is finitely generated (in general only $\subseteq$ holds)*

*Proof.* Evidently $S^{-1}I$ is again a fractional ideal of $S^{-1}A$.

a) Clear because $\frac{i+j}{s} = \frac{i}{s} + \frac{j}{s}$.
b) We have $\frac{ij}{s} = \left(\frac{i}{s}\right)\left(\frac{j}{1}\right)$ whence $S^{-1}(IJ) \subseteq (S^{-1}I)(S^{-1}J)$. Conversely we have $\left(\frac{i}{s}\right)\left(\frac{j}{t}\right) = \frac{ij}{st}$ so we ahve the reverse inclusion.
c) Suppose that $\frac{x}{s} \in S^{-1}(I:J)$ and $\frac{j}{t} \in S^{-1}J$. Then $\left(\frac{x}{s}\right)\left(\frac{j}{t}\right) = \frac{xj}{st} \in S^{-1}I$ because $x \in (I:J)$. Therefore $S^{-1}(I:J) \subseteq (S^{-1}I : S^{-1}J)$. Conversely suppose $J = (j_1, ..., j_n)$ and let $x \in (S^{-1}I : S^{-1}J)$. Then for each $k$ we have $x\left(\frac{j_k}{1}\right) = \frac{i_k}{s_k}$ with $i_k \in I$ and $s_k \in S$. Put $s := s_1 ... s_n \in S$, so $sxj_k \in I$ for every $k$, whence $sxJ \subseteq I$ as the $j_k$ generate $J$, that is $sx \in (I:J)$. Therefore $x = \frac{sx}{s} \in S^{-1}(I:J)$ as required. (Finite generation is what lets a single $s$ serve all of $J$.)

■

**Lemma 3.29.10**
*Let $A$ be an integral domain contained in its quotient field $K$. Suppose $I$ is a fractional ideal then*

$$I = \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$$

*In particular we have*

$$A = \bigcap_{\mathfrak{m}} A_{\mathfrak{m}}$$

*Proof.* Evidently $I \subseteq \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Suppose that $x \in \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Consider the ideal $\mathfrak{a} := \{a \in A \mid ax \in I\}$. Suppose $\mathfrak{a}$ is proper then by (3.3.15) it is contained in a maximal ideal $\mathfrak{m}$. As $x \in I_{\mathfrak{m}}$ we have $(A \setminus \mathfrak{m}) \cap \mathfrak{a} \neq \emptyset$ and $\mathfrak{a} \subseteq \mathfrak{m}$ a contradiction. Therefore $\mathfrak{a} = A$ and in particular $x = 1.x \in I$ as required. ■

**Proposition 3.29.11 (Invertibility is a local property)**
*Let $A$ be an integral domain with quotient field $K$, and $I$ a fractional ideal of $A$. Consider the conditions*

*a)* *$I$ is invertible*
*b)* *$I_{\mathfrak{p}}$ is invertible for all prime ideals $\mathfrak{p}$*
*c)* *$I_{\mathfrak{m}}$ is invertible for all maximal ideals $\mathfrak{m}$*

*Then a) $\implies$ b) $\implies$ c), and if $I$ is finitely generated then c) $\implies$ a). In particular all three are equivalent when $A$ is Noetherian, as then every fractional ideal is finitely generated (3.29.2).*

*Proof.* a) $\implies$ b) Let $IJ = A$ for a fractional ideal $J$. By (3.29.9) $I_{\mathfrak{p}}J_{\mathfrak{p}} = (IJ)_{\mathfrak{p}} = A_{\mathfrak{p}}$, so $I_{\mathfrak{p}}$ is invertible with inverse $J_{\mathfrak{p}}$.

b) $\implies$ c) Every maximal ideal is prime (3.3.60) so we are done.

c) $\implies$ a) Suppose that $I$ is finitely generated. Then

$$\begin{aligned}
I(A:I) &= \bigcap_{\mathfrak{m}} (I(A:I))_{\mathfrak{m}} && (3.29.10)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A:I)_{\mathfrak{m}} && (3.29.9)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A_{\mathfrak{m}} : I_{\mathfrak{m}}) && (3.29.9)\\
&= \bigcap_{\mathfrak{m}} A_{\mathfrak{m}} && (3.29.6)\\
&= A && (3.29.10)
\end{aligned}$$

where the third equality uses that $I$ is finitely generated, and the second last that $I_{\mathfrak{m}}$ is invertible by assumption. ■

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